JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to VICTORY AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

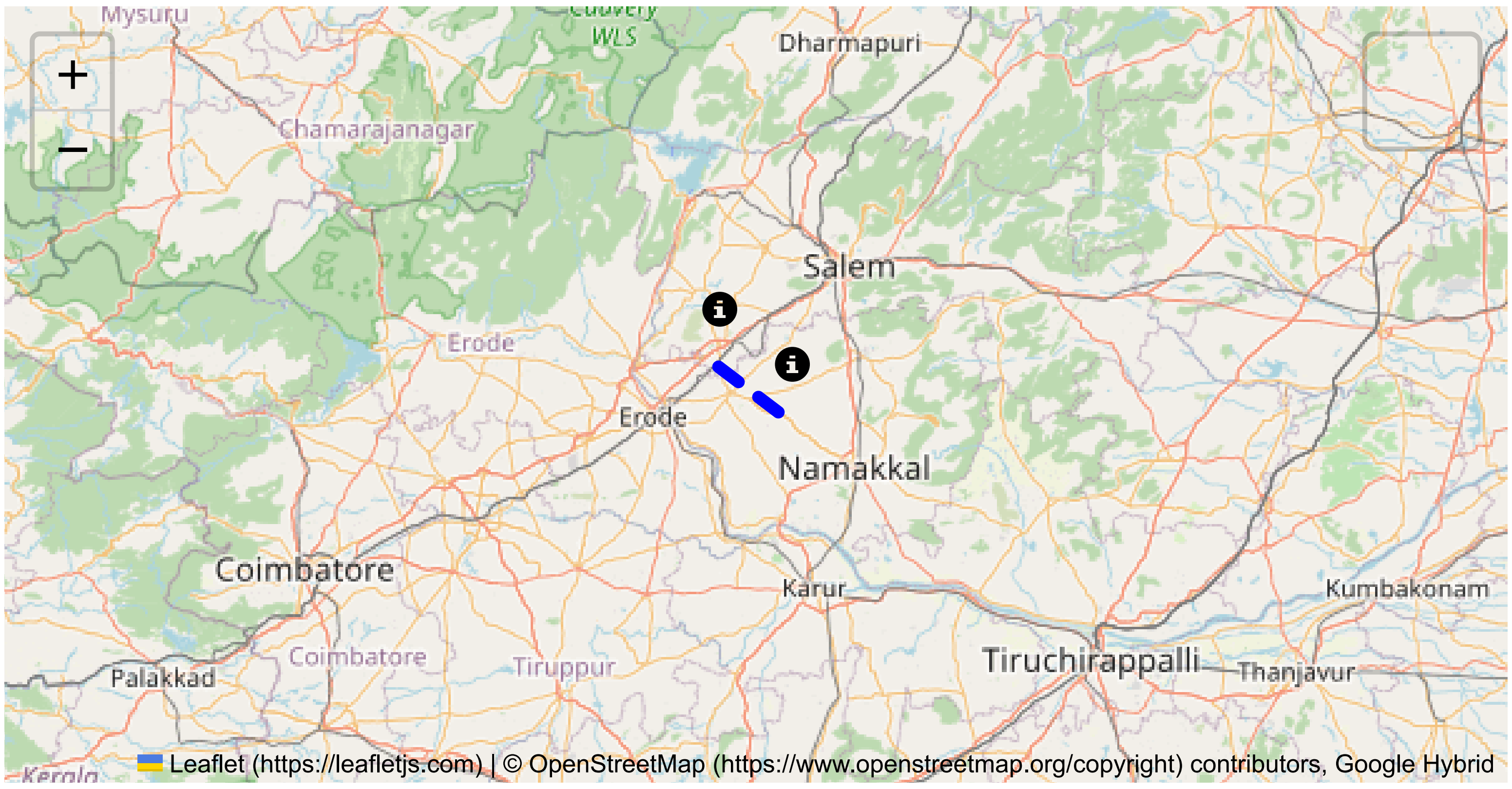
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

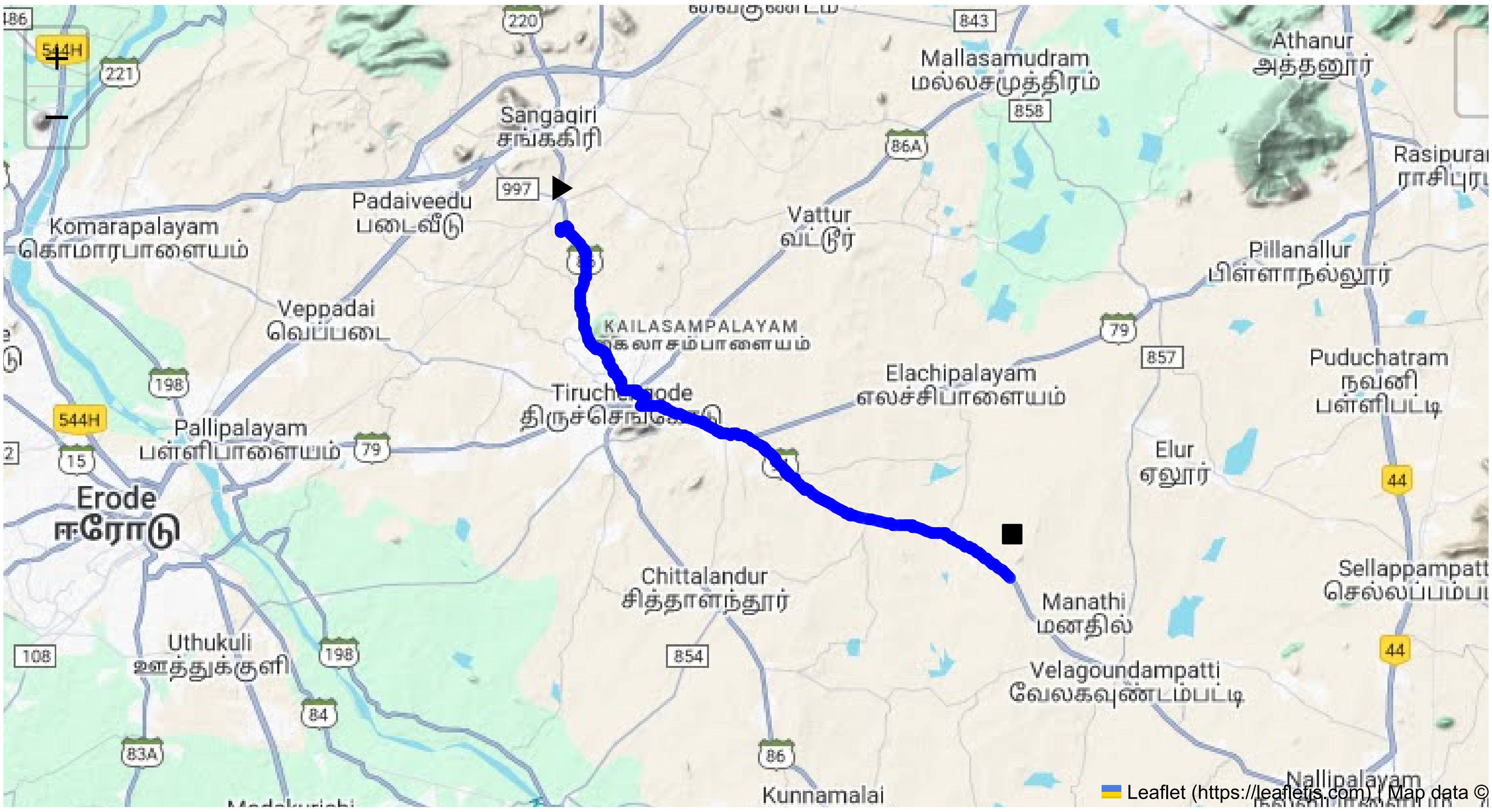
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 24.89 km
Estimated Duration: 0.6 hours
Adjusted Duration (Heavy Vehicle): 0.7 hours
Start: (11.4381, 77.8734)
End: (11.319195, 78.031005)



Welcome to the Journey Risk Management Study

To provide a comprehensive analysis, here is a detailed breakdown of the route from CVQF+23W, Sangagiri, Tamil Nadu 637301, India to 829J+MC Koothampoondi, Tamil Nadu, India:

1. Overview of the Route Map

- **Distance**: Approximately 24.89 kilometers.

- **Estimated Travel Time**: Around 35-40 minutes for heavy vehicles.
- **Key Roads**: The journey primarily involves local roads with limited highway exposure.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather Patterns**: Sangagiri and Koothampoondi experience a tropical climate with three major seasons: summer (March to June), monsoon (July to October), and winter (November to February).
- **Hazards**: The monsoon season may bring heavy rainfall leading to waterlogged roads and reduced visibility. Slippery conditions are prevalent during this period.

3. Analysis of Traffic Patterns

- **Peak Traffic Hours**: Morning hours (7 AM to 9 AM) and evening hours (5 PM to 7 PM) tend to see increased traffic due to commuting. Traffic is generally lighter during midday and late evening.
- **Congestion-Prone Areas**: Larger towns or markets along the route may experience congestion, particularly near schools and marketplaces.

4. Assessment of Road Quality and Infrastructure

- **Road Condition**: Generally, roads in this region may have uneven patches and occasional potholes, especially after monsoon rains. Limited street lighting in rural segments may affect visibility at night.
- **Infrastructure**: The route lacks major expressway features, and existing road infrastructure may not fully support heavy vehicles.

5. Suggestions for Alternative Routes for Emergencies

- **Diversion Options**: Smaller rural roads connecting nearby villages may serve as alternate routes; however, they may not be well-suited for heavy vehicles or be properly maintained.
- **Emergency Shelters**: Identifying locations such as schools or public buildings that can serve as temporary shelters is crucial.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Transport Regulations**: Movement of hazardous materials typically requires adherence to strict guidelines, including proper labeling, necessary permits, and restricted transit times.
- **Local Restrictions**: Be aware of regional check-posts where documents may be verified by authorities.

7. Overview of Historical Incidents

- **Incident Records**: Historical data on accidents involving heavy vehicles or hazardous materials is limited. However, vigilance is required on local roads due to sharp turns and lack of barriers.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Zones**: Agricultural fields and water bodies dot the landscape. Special care must be taken not to spill materials that could impact local ecology.
- **Wildlife**: The area is not known for significant wildlife crossings but caution is advised near forested areas.

9. Analysis of Communication Coverage

- **Network Coverage**: Mobile network coverage is generally reliable in more developed areas but may be spotty in rural segments. Clear communication is critical for coordination.

10. Estimated Emergency Response Times

- **Response Time**: Urban segments might see faster emergency response times (~20-30 minutes), while rural areas could experience delays up to an hour, depending on severity and location.

11. Overall Summary of Risk Assessment

- **Risks Identified**: Potential hazards due to weather conditions, poor road quality, and limited infrastructure. Traffic congestion during peak hours can delay transportation; regulations need to be strictly followed to prevent legal issues.
- **Recommendations**: Drivers should be prepared for diverse conditions, maintaining caution and ensuring that emergency protocols are well understood and communicated.

This risk assessment provides a detailed overlook of potential hazards and measures to ensure safe transit. Emphasizing adherence to regulations and preparedness for sudden changes in conditions is crucial for minimizing risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	Medium	11.43971, 77.87348	30 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	High	11.38346, 77.89975	15 KM/Hr
6	Turn	Medium	11.38190, 77.90254	30 KM/Hr
7	Turn	Medium	11.38186, 77.90258	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
8	Turn	Medium	11.38182, 77.90259	30 KM/Hr
9	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
8	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
7	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38346, 77.89975

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38190, 77.90254

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38186, 77.90258

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38182, 77.90259

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.37841, 77.90140